

Malware and Endpoint Security

Understand malicious software and protect the devices people use

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Opening question

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A laptop suddenly becomes slow and starts opening unfamiliar windows. What evidence would help distinguish malware from an ordinary technical problem?



By the end, you will be able to:

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1. Recognize major malware categories without focusing on malicious implementation.

2. Explain common infection paths and endpoint protections.

3. Choose safe first actions when a device may be compromised.



1. Recognize malware behavior

Malware is identified by its malicious purpose and actions. Its name, appearance, and visible symptoms can be misleading.

An innocent name does not guarantee safety

A malware is a software intended to cause harm



Malicious software may steal information, disrupt services, spy on users, gain control, or enable later attacks.

1 Some malware hides quietly.

2 A symptom alone does not prove infection.

Viruses attach and spread through user action

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Sality: a file-infecting virus

Detected item	Alert level	Date	Action taken
Virus:Win32/Sality.AT	Severe	09-Aug-15 7:08 PM	Cleaned
Virus:Win32/Sality.AT	Severe	09-Aug-15 7:07 PM	Cleaned
Virus:Win32/Sality.AT	Severe	09-Aug-15 7:06 PM	Cleaned
Virus:Win32/Sality.AT	Severe	09-Aug-15 7:05 PM	Cleaned
Virus:Win32/Sality.AT	Severe	09-Aug-15 7:04 PM	Cleaned
Virus:Win32/Sality.AT	Severe	09-Aug-15 7:03 PM	Cleaned
Virus:Win32/Sality.AT	Severe	09-Aug-15 7:02 PM	Cleaned
Virus:Win32/Sality.AT	Severe	09-Aug-15 7:01 PM	Cleaned
Virus:Win32/Sality.AT	Severe	09-Aug-15 7:00 PM	Cleaned
Virus:Win32/Sality.AT	Severe	09-Aug-15 6:58 PM	Cleaned
Virus:Win32/Sality.AT	Severe	09-Aug-15 6:57 PM	Cleaned
Virus:Win32/Sality.AT	Severe	09-Aug-15 6:55 PM	Cleaned
Virus:Win32/Sality.AT	Severe	09-Aug-15 6:54 PM	Cleaned
Virus:Win32/Sality.AT	Severe	09-Aug-15 6:53 PM	Cleaned

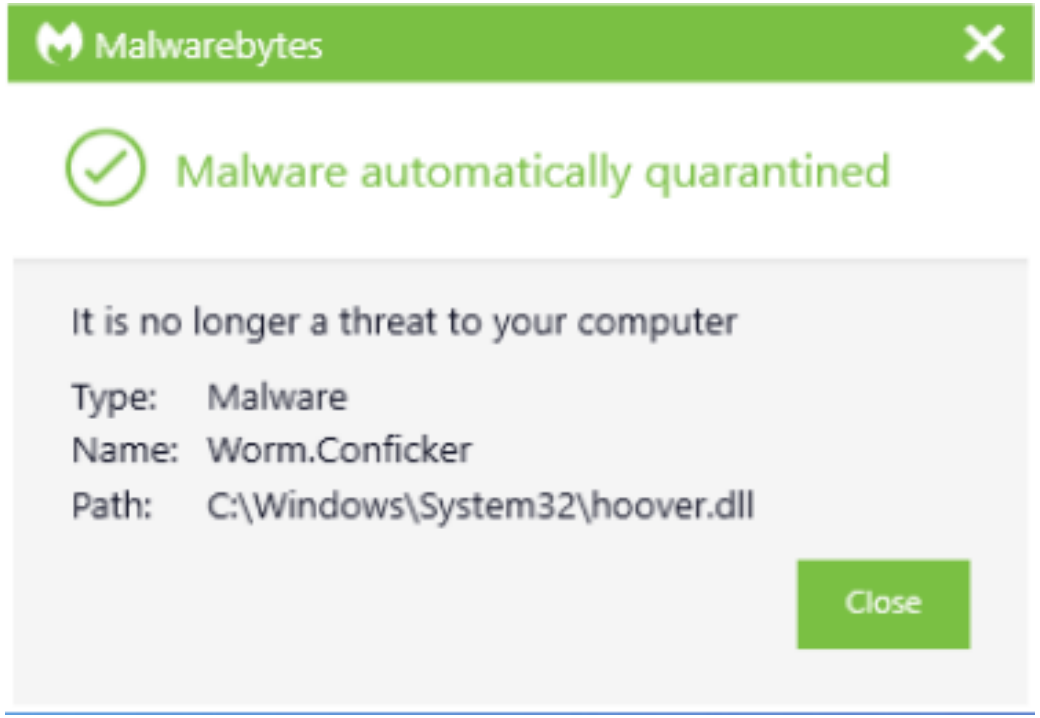
Windows Defender detection history (detail). The alert is visible; the infected program can look ordinary.

A virus typically becomes active when an infected file or program is executed and may copy itself into other content.

1. Sharing infected files can spread the problem.
2. Modern threats often combine several techniques.

Worms spread through reachable systems

Conficker: a network worm



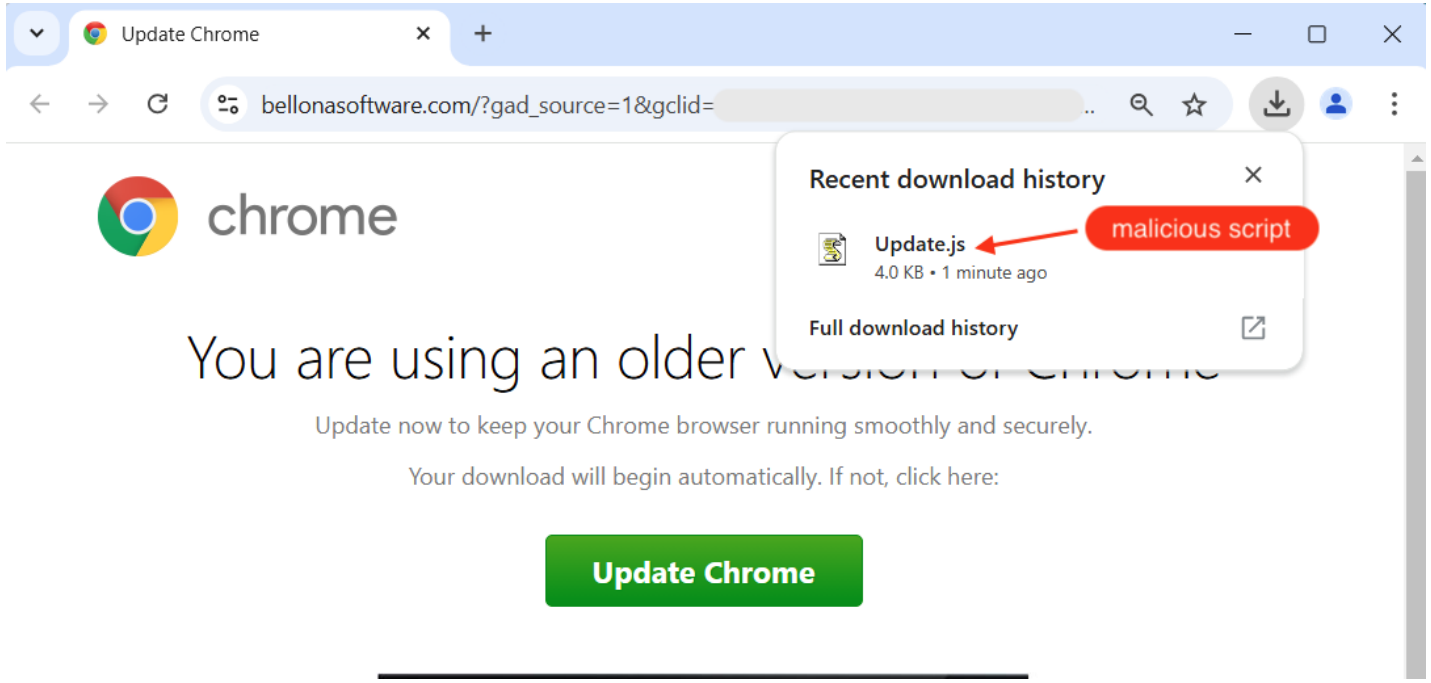
A protection alert names Worm.Conficker. The worm itself does not need to display a window.

A worm can propagate across networks by exploiting weaknesses without waiting for every user to open a file.

1. Segmentation can limit spread.
2. Fast patching reduces exploitable paths.

Trojans pretend to be useful

SocGholish: a fake browser update



The page imitates Chrome. The downloaded “Update.js” is a malicious script, not a browser update.

A malicious program may be disguised as a game, update, document, utility, or pirated application.

1. Download only from trusted sources.
2. Appearance and filename do not prove safety.

Ransomware disrupts and extorts

WannaCry: an actual ransom screen



A payment demand and countdown appear after encryption. This historical example also spread as a worm.

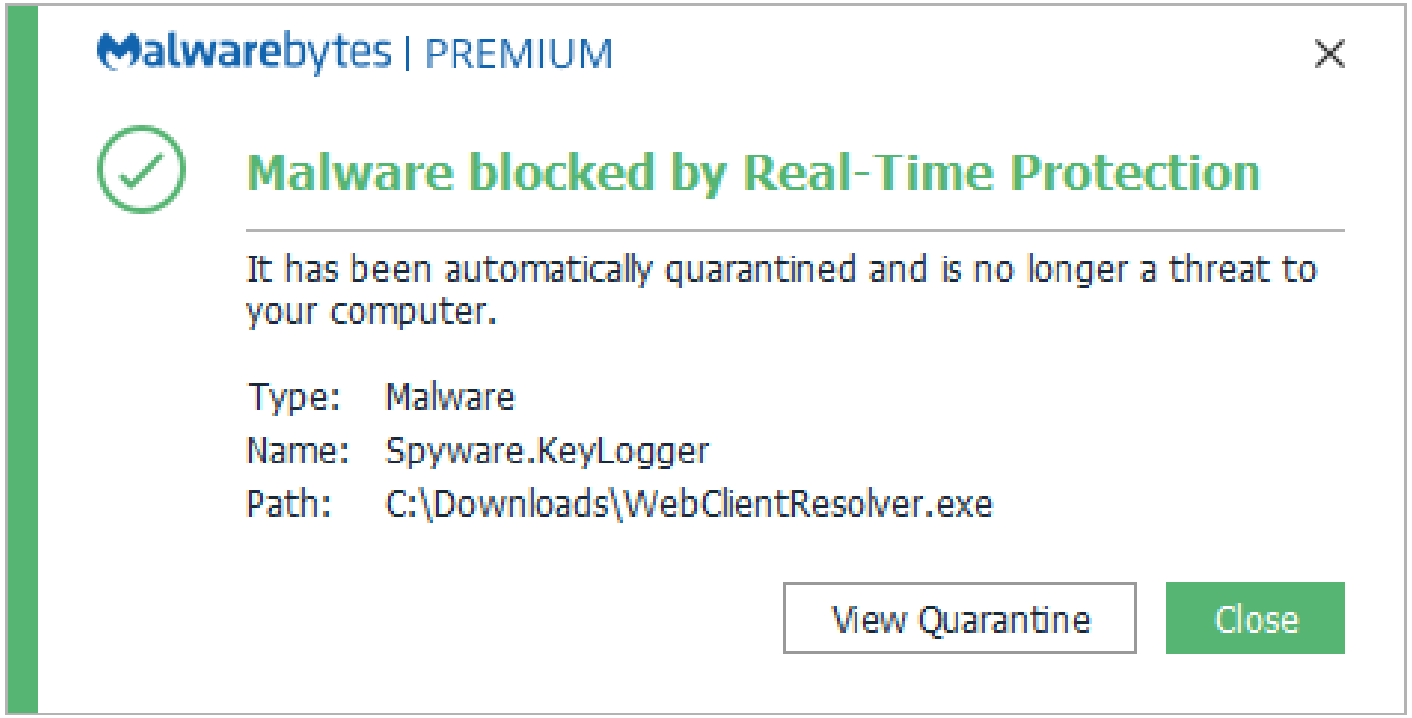
Ransomware commonly encrypts or blocks systems and may also steal data to increase pressure.

1. Payment does not guarantee recovery.
2. Backups and response preparation reduce impact.

Spyware collects information quietly

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Spyware.KeyLogger: a protection alert



The security tool reports a blocked keylogger. Spyware often works without a visible sign.

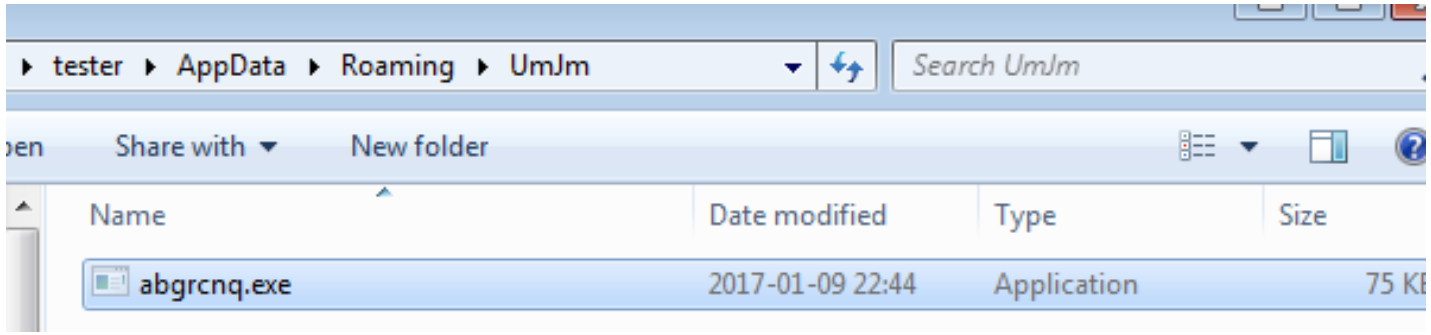
Spyware collects information about the user or the device. A keylogger can capture keystrokes, including sensitive information.

1. Hidden activity may consume data and resources.
2. Monitoring and endpoint protection help reveal it.

Bots put a device under external control

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Neutrino Bot: an installed file



Bots let an external operator use the device for spam, fraud, attacks, or further access.

Most users will not see a special “bot” screen.

Researchers found this ordinary-looking application in AppData.

1. Hidden activity may consume data and resources.
2. Monitoring and endpoint protection help reveal it.

Quick check

Can a computer be infected even when no warning window appears? Explain.



Activity

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Look at 8 examples of suspicious computer behavior. For each one, guess the most likely type of malware and say what more information you would need to be sure.

Activity: cases 1–4

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Case	What you observe
1	An infected program is shared on a USB drive. Other executable files then become infected.
2	Several unpatched computers become infected without each user opening a file.
3	A “free game” installs an unwanted remote-control program.
4	Course files no longer open. A screen demands payment to restore them.

Activity: cases 5–8

Case	What you observe
5	A program secretly records what a student types.
6	A device sends attack traffic when a remote operator tells it to.
7	A laptop is slow, but no scan, process, or network evidence is available.
8	A fake update steals information and downloads ransomware.



2. Break the infection path

Malware often depends on unsafe content, weak configuration, excessive privilege, or unpatched software.

Malicious messages deliver content

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Attachments, links, shared documents, and fake login pages can lead users toward harmful downloads or credential theft.

- Verify unexpected content.
- Automatic filtering and user reporting provide extra protection against malicious messages

Untrusted downloads disguise risk

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Pirated software, unofficial app stores, fake updates, and bundled installers may include unwanted or malicious components.

1 Use approved sources and software lists.

2 Check publisher and update path.

Unpatched vulnerabilities create openings

Attackers can exploit known software flaws when updates are delayed or unsupported software remains in use.

- Prioritize internet-facing and critical systems.
- Replace software that no longer receives fixes.

Excess privilege increases damage

Malware running with administrator access can change more settings, reach more data, and disable more controls.

- Use standard accounts for daily work.
- Approve elevated actions carefully.

Removable media crosses boundaries



Unknown or infected USB devices can introduce malicious files or unauthorized hardware into protected environments.

1. Use organizational media policy.
2. Do not connect found or unknown devices.

Weak configuration helps persistence

Automatic execution, exposed services, default credentials, and disabled protections can make infection and continued access easier.

- Use secure baselines.
- Monitor for unexpected configuration changes.

Supply-chain trust needs verification



Legitimate software updates, packages, or vendors can become a path when their development or delivery process is compromised.

1 Use trusted signed sources.

2 Track dependencies and supplier notices.

Quick check

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Why can an attachment from a known contact still be dangerous?



Activity



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Trace four infection stories from initial contact to execution, privilege, persistence, and impact. Place possible controls at each story.

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Story	Initial contact	Execution	Privilege	Persistence	Impact	Possible control
1. Malicious email attachment	User receives an infected document	User opens the attachment				
2. Fake software / fake update	User downloads a fake game or update	User runs the installer				
3. Network worm	Vulnerable computer is reachable on the network	Worm exploits an unpatched service				
4. Infected USB	User connects a USB with an infected program	User runs the infected file				



3. Protect and respond at the endpoint

Secure configuration, updates, least privilege, detection, backups, and fast reporting work together.

Updates remove known weaknesses

Operating systems, browsers, applications, firmware, and security tools should receive supported updates through trusted mechanisms.

- Enable automatic updates where appropriate.
- Track devices that cannot update.

Endpoint protection watches behavior

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Antivirus and endpoint tools can block known threats and identify suspicious files, processes, settings, or communications.

1 Detection quality changes over time.

2 Alerts need investigation and response.

Secure configuration reduces exposure

Disable unnecessary services, restrict installation, lock screens, encrypt storage, and use approved software.

- Defaults may prioritize convenience: the settings that come with a device may be easy to use, but not always the most secure
- Configuration should be repeatable and reviewable: Every computer on the organization level should follow the same security rules

Backups reduce ransomware impact



Protected, separated, and tested backups can restore important information without relying on the attacker.

1 A connected backup may also be encrypted.

2 Test recovery before an incident.

Suspicious devices should be reported quickly

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If you suspect infection: stop sensitive work, avoid making unnecessary changes, and report it quickly so trained staff can decide what to do next.

- Do not continue experimenting on the device: don’t keep clicking around, opening files, running scans randomly, or trying to “see what happens.
- Follow guidance before disconnecting critical systems: for important machines like servers, lab systems, don’t just shut them down immediately. IT/security staff may need to preserve evidence

Recovery must return to a trusted state



Recovery is complete only when the system is clean, the weakness is fixed, credentials are safe, data is restored, and the system is monitored afterward

1. Restoring files alone may restore the weakness: Restore clean data, but also fix the cause of the infection
2. Document cause and preventive improvement:

Quick check

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Why is antivirus useful but insufficient as the only endpoint control?



Use six questions before recommending a design

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#	Question to consider
1	Observed symptom
2	Possible infection path
3	Privilege level
4	Affected data
5	Safe first action
6	Recovery evidence

Activity: a suspicious library computer

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- **A fake update was installed**
- **The browser opens unknown pages**
- **A student used email after the installation**

Identify possible malware behavior, immediate reporting and containment steps, affected accounts, and controls that should prevent recurrence.

Five ideas to keep after today

- Malware categories describe different spreading methods and harmful goals.
- Messages, downloads, vulnerabilities, media, privileges, and suppliers can become infection paths.
- Updates, secure configuration, least privilege, and endpoint protection reduce opportunity.
- Protected and tested backups reduce the impact of ransomware and device failure.
- Fast reporting and trusted recovery matter more than guessing the exact malware name.

Q & A· Questions 1–5

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01 What is malware?

02 How does a virus commonly spread?

03 How is a worm different from a virus?

04 What is a Trojan?

05 What is ransomware?

Q & A· Answers 1–5

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1. Software intentionally designed to cause harm, steal information, disrupt service, or enable unauthorized control.

2. Through execution and sharing of infected files or programs.

3. A worm can propagate through networks without depending on every user to open an infected file.

4. Malware disguised as legitimate or useful software or content.

5. Malware that blocks or encrypts systems or data to demand payment, often combined with data theft.

Q&A review · Questions 6–10

06 Why do updates matter?

07 Why does least privilege help?

08 Why is antivirus not enough?

09 What should a user do after noticing suspicious behavior?

10 What makes a backup useful after malware?

Q & A· Answers 6–10

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6. They correct known vulnerabilities that malware and attackers may exploit.

7. It limits what malicious code can change or access.

8. No tool detects every threat, and protection also depends on configuration, updates, access, backups, and users.

9. Stop sensitive activity and report through the approved support process.

10. It is protected from the incident, recent enough, and tested for successful restoration.

Picture sources and further reading

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Virus	https://learn.microsoft.com/en-us/answers/questions/2668420/windows-defender-is-taking-action-to-clean-detecte
Worm	https://www.malwarebytes.com/blog/detections/worm-conficker
Trojan	https://www.malwarebytes.com/blog/threat-intel/2024/12/malicious-ad-distributes-socgholish-malware-to-kaiser-permanente-employees
Ransomware	https://www.microsoft.com/en-us/security/blog/2017/05/12/wannacrypt-ransomware-worm-targets-out-of-date-systems/
Spyware	https://www.malwarebytes.com/blog/detections/spyware-keylogger
Bot	https://www.malwarebytes.com/blog/news/2017/01/post-holiday-spam-campaign-delivers-neutrino-bot
Prevention	https://www.cisa.gov/stopransomware

Assignment - Trace a Malware Infection

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1. Choose ONE scenario:

- Malicious email attachment
- Fake software/update
- Network worm
- Infected USB

2. Your Task

Create **one attack story** using this path:

Initial Contact → Execution → Privilege → Persistence → Impact → Recovery

Next Week: Networking Fundamentals for Security